Nathan DSilva

University of Waterloo - Combinatorics & Optimization

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Work Experience

Lead Infrastructure Developer, Nexa Ligo Inc. Jun 2025 - Present

- Patented a novel proxy service that provides a runtime for browser extensions exclusively through a web portal.
- Architected and audited GDPR and CCPA compliance for 21 tenant businesses' data-handling pipelines. Identified and closed 6+ critical security risks across authentication services.
- Maintained and improved infrastructure as code (Terraform), ensuring scalability and uptime of services including CloudFront, Lambda, ECS, EKS, and RDS.

Lead Developer, University of Waterloo - Games Institute May 2024 - Dec 2024

- Primary technical lead for the project, responsible for architecture, design, and implementation of new features.
- Independently led teams of 4-7 developers to overhaul the full-stack application written in Go and React.js.
- Pushed project timelines forward by 6 months by significantly improving development speed and code quality.

Machine Learning Engineer (Part-time), University of Waterloo Sep 2024 - Dec 2024

- Improved ECG diagnosis accuracy from 86.1% to 98.6% while simultaneously reducing model size by a factor of 250 times by architecting a neural network from scratch with PyTorch.
- Co-authored a research paper on utilizing machine-learning and 5G cellular networks for reducing patient diagnosis time.

Project Lead, Scrawlr Development Inc. Jan 2023 - Aug 2023

- Lead developer on over 20 projects, including a reactivity library for Vue.js, a browser extension plugin for Vite, and various API specification & mocking utilities.
- Mentored several junior developers by providing technical guidance, conducting code reviews, and completing performance evaluations.
- Collaborated across 4 different teams to develop tools, infrastructure, CI/CD, web components, and APIs.

Internal Tools Developer, Scrawlr Development Inc. May 2022 - Dec 2022

Junior Network Administrator, Questenterprise Inc. Sep 2021 - Dec 2021

Personal Projects

Infinite Craft Wiki
Web Interface Vue, Tailwind, Data handling

Used by over 84k players, compiled 6.6 million recipes from the game "Infinite Craft" into a searchable interface without requiring a database, and achieved 1.1 million page views in the first 4 months of operation.

MineNewt
Machine Learning Rust, Python, Neural Networks

Created a machine learning library from scratch in Rust, and researched novel encodings of neural networks with stochastic (probabilistic) bits. Developed a Python library for simulating the performance of networks encoded this way, and embedded it into a custom chip architecture.

— See GitHub for more projects —

Skills

Frontend & Web Development - Javascript/Typescript · React · Vue · Tailwind · REST APIs · PHP · HTML · CSS
Programming Languages - Python · Rust · Node.js · Go · C · C++ · Bash · SQL · Java · C#
Infrastructure & Tools - Docker · AWS · PostgreSQL · MongoDB · MySQL · Git · Linux · CircleCI
Soft Skills - Delegative Leadership · Technical Mentoring · Conflict Resolution · Automation

Education

University of Waterloo 2020 - 2025
Computational Mathematics, Combinatorics & Optimization (Co-op)